

Week 3: The Avellaneda-Stoikov Model

A worked example of optimal market making via stochastic control

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Learning Objectives

- ▶ Set up the Avellaneda-Stoikov control problem with CARA utility
- ▶ Understand the HJB equation and its reduction via the ansatz
- ▶ Interpret the reservation price and optimal half-spread

Lecture 1: Setup and Dynamics

- ▶ Mid-price as arithmetic Brownian motion (chosen for tractability)
- ▶ Poisson order flow with intensity $\lambda(d) = A \exp(-\kappa d)$
- ▶ Wealth and inventory dynamics under continuous quoting
- ▶ Exponential (CARA) utility decouples value from wealth

Lecture 2: HJB and Closed-Form Quotes

- ▶ Dynamic programming principle $\rightarrow$ HJB equation
- ▶ Separation ansatz $V = -\exp(-\gamma(x+qS)) \theta(t,q)$
- ▶ Reservation price $r = S - q \gamma \sigma^2 (T-t)$
- ▶ Half-spread $= (1/\gamma) \ln(1+\gamma/\kappa) + (\gamma \sigma^2 / 2)(T-t)$

- ▶ Implement the closed-form quotes in PyTorch
- ▶ Visualize reservation price and spread vs. inventory
- ▶ Run a risk-aversion (γ) sensitivity sweep

Reading & Problem Set 3 (Python/PyTorch)

- ▶ Avellaneda-Stoikov (2008); Cartea-Jaikumar-Penalva Ch. 10
- ▶ PS3: ODE solver, closed-form quotes, simulation, gamma sweep